Física Nuclear y Subnuclear – Tarea 3

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Problema 1

La amplitud de dispersión, en la primera aproximación de Born, está dada por

$$f(\theta,\phi) = -\frac{m}{2\pi\hbar^2}\int e^{i(\vec{k}-\vec{k'})\cdot\vec{r_0}}V(\vec{r_0})d^3r_0,$$

donde $k = k\hat{r}$ y $k' = k\hat{z}$ son vectores de onda con la misma magnitud, pero apuntando en la dirección de la trayectoria final e inicial de la partícula respectivamente (siendo θ y ϕ las coordenadas angulares de $\hat{r}$).

a) Muestra que para un potencial central $V(\vec{r}) = V(r)$ la expresión para la amplitud de dispersión se simplifica a

$$f(\theta) = -\frac{2m}{\hbar^2\kappa}\int_0^\infty r_0 V(r_0)\sin(\kappa r_0)dr_0,$$

con $\vec{\kappa} = \vec{k} - \vec{k'}$.

a) $f(\theta,\phi) = \frac{-m}{2\pi\hbar^2}\int e^{i(\vec{k}-\vec{k}')\cdot\vec{r}_0} V(\vec{r}_0)\, d^3r_0$

• con $V(\vec{r}_0) = V(r)$:

Dado que $|\vec{k}| = |\vec{k}'| = k$

$$\kappa = |\vec{k}-\vec{k}'| = \sqrt{(\vec{k}-\vec{k}')\cdot(\vec{k}-\vec{k}')} = \sqrt{k^2+k^2-2k^2\cos\theta} = \sqrt{2k^2(1-\cos\theta)} = 2k\,\text{sen}\left(\frac{\theta}{2}\right)$$



$$f(\theta,\phi) = \frac{-m}{2\pi\hbar^2}\int e^{i\vec{K}\cdot\vec{r}_0} V(r_0)\, d^3r_0$$

↳ $d^3r_0 = r_0^2\,\text{sen}\theta\, dr_0 d\theta_0 d\phi_0$

$$= \frac{-m}{2\pi\hbar}\int_0^{2\pi} d\phi_0 \int_0^{\pi} d\theta\,\text{sen}\theta \int_0^{\infty} r_0^2\, dr_0\, d\phi_0\, e^{i\vec{K}\cdot\vec{r}_0} V(r_0)$$

$$\vec{K}\cdot\vec{r_0} = Kr_0\cos\theta_0$$

$$= \frac{-m}{\hbar}\int_0^{\infty} r_0^2 \int_0^{\pi} d\theta\,\text{sen}\theta\, e^{i(Kr_0\cos\theta_0)} V(r_0)\, dr_0$$

$$\int_0^{\pi} d\theta\,\text{sen}\theta_0\, e^{iKr_0\cos\theta_0}$$

$u = \cos\theta_0 \; ; Kr_0$

$\underline{du} = -\text{sen}\theta_0\, d\theta$

iKr_0

$$-\int \frac{du}{iKr_0} e^{iKu} = -\frac{e^{iKu}}{iKr_0}\Big|_{U_0}^{U_F} = -\left[e^{iKr_0\cos\theta_0}\right]_0^{\pi} = -\frac{\left[e^{-iKr_0} - e^{iKr_0}\right]}{iKr_0}$$

$$= \frac{2i\,\text{sen}(Kr_0)}{iKr_0} = \frac{2\,\text{sen}(Kr_0)}{Kr_0}$$

$$f(\theta,\phi) = \frac{-m}{\hbar}\int_0^{\infty} r_0^2\,\frac{2\,\text{sen}(Kr_0)}{Kr_0} V(r_0)\, dr_0$$

$$f(\theta) = \frac{-2m}{K\hbar}\int_0^{\infty} r_0\,\text{sen}(Kr_0) V(r_0)\, dr_0$$

b) Muestra que para el potencial de Yukawa, dado por

$$V(r) = \beta \frac{e^{-\mu r}}{r},$$

donde $\mu > 0$ y β son constantes, la primera aproximación de Born arroja que

$$f(\theta) = -\frac{2m\beta}{\hbar(\mu^2 + \kappa^2)}.$$

$$f(\theta) = \frac{-2m}{\kappa\hbar} \int_0^\infty r_0 \operatorname{sen}(\kappa r_0)\, V(r_0)\, dr_0$$

con $V(r) = \beta \dfrac{e^{-\mu r}}{r}$

$$f(\theta) = \frac{-2m}{\kappa\hbar} \int_0^\infty r_0 \operatorname{sen}(\kappa r_0)\, \beta \frac{e^{-\mu r_0}}{r_0}\, dr_0$$

$$f(\theta) = -\frac{\beta 2m}{\kappa\hbar} \int_0^\infty \operatorname{sen}(\kappa r_0)\, e^{-\mu r_0}\, dr_0$$

$u = \operatorname{sen}(\kappa r_0) \qquad dv = e^{-\mu r_0} dr_0$

$du = \cos(\kappa r_0)\kappa\, dr_0 \qquad v = -\frac{1}{\mu} e^{-\mu r_0}$

$$f(\theta) = -\frac{\beta 2m}{\kappa\hbar} \left(\operatorname{sen}(\kappa r_0) \cdot -\frac{1}{\mu} e^{-\mu r_0} \Big|_0^\infty - \int -\frac{1}{\mu} e^{-\mu r_0} \kappa \cos(\kappa r_0) \right)$$

$$-\frac{1}{\mu} \int e^{-\mu r_0} \kappa \cos(\kappa r_0)\, dr_0$$

$u = \cos(\kappa r_0)\kappa \qquad dv = e^{-\mu r_0} dr_0$

$du = -\operatorname{sen}(\kappa r_0)\kappa^2 dr_0 \qquad v = -\frac{1}{\mu} e^{-\mu r_0}$

$$f(\theta) = -\frac{\beta 2m}{\kappa\hbar} - \frac{1}{\mu} \left[\cos(\kappa r_0)\kappa \left(-\frac{1}{\mu}\right) e^{-\mu r_0} \Big|_0^\infty - \int_0^\infty \left(-\frac{1}{\mu} e^{-\mu r_0}\right)\left(-\operatorname{sen}(\kappa r_0)\kappa^2 dr_0\right) \right]$$

$$f(\theta) = -\frac{\beta 2m}{\kappa\hbar} \cdot \left(-\frac{\kappa}{\mu^2} + \frac{\kappa^2}{\mu^2} \int_0^\infty \operatorname{sen}(\kappa r_0)\, e^{-\mu r_0}\, dr_0 \right)$$

$$= -\frac{\beta 2m}{\kappa\hbar} \cdot -\left(\frac{-\kappa}{\mu^2 + k^2} \right) = -\frac{2m\beta}{\kappa\hbar} \cdot \frac{\kappa}{(\mu^2 + k^2)}$$

$$\int_0^\infty \operatorname{sen}(\kappa r_0)\, e^{-\mu r_0}\, dr_0 = -\frac{1}{\mu} - \frac{\kappa^2}{\mu} \int_0^\infty \operatorname{sen}(\kappa r_0)\, e^{-\mu r_0}\, dr_0$$

$$\left(1 + \frac{k^2}{\mu^2}\right) \int (\ldots) = -\frac{\kappa}{\mu^2}$$

$$\int (\ldots = \frac{-\kappa}{\mu^2 + k^2}$$

$$f(\theta) = \frac{-2m\beta}{\hbar} \cdot \frac{1}{(\mu^2 + k^2)}$$

c) Toma el límite de $\mu \to 0$ en el resultado de b), y sustituyendo el valor apropiado para β concluye que la sección eficaz diferencial para el potencial de Coulomb (dispersión de Rutherford) es

$$\frac{d\sigma}{d\Omega} = \left(\frac{q_1 q_2}{16\pi\epsilon_0 E \sin^2(\theta/2)} \right)^2 .$$

$$f(\theta) = \frac{-2m\beta}{\hbar} \cdot \frac{1}{(\mu^2 + K^2)} \xrightarrow{\mu \to 0} f(\theta) = \frac{-2m\beta}{\hbar} \cdot \frac{1}{K^2}$$

$$V(r) = \beta \frac{e^{-\mu r}}{r} \Rightarrow V(r) = \frac{\beta}{r} \quad \text{(coulombiano)}$$

$$\therefore \beta = \frac{q_1 q_2}{4\pi\varepsilon_0}$$

$$\Rightarrow f(\theta) = -\frac{2m q_1 q_2}{4\pi\varepsilon_0 \hbar} \cdot \frac{1}{K^2}$$

de a) $K = 2k \operatorname{sen}\left(\frac{\theta}{2}\right) \Rightarrow K^2 = 4k^2 \operatorname{sen}^2(\theta/2)$

$k = \frac{\sqrt{2mE}}{\hbar} \Rightarrow k^2 = \frac{2mE}{\hbar^2}$

$$\therefore f(\theta) = \frac{-2m q_1 q_2 \hbar^2}{4\pi\varepsilon_0 \hbar \, 4\, 2mE \operatorname{sen}(\theta/2)} = \frac{-q_1 q_2 \hbar}{16\pi\varepsilon_0 E \operatorname{sen}(\theta/2)}$$

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2 = \left(\frac{q_1 q_2 \hbar}{16\pi\varepsilon_0 E \operatorname{sen}(\theta/2)} \right)^2$$